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Unit 3 : Session 9 : SLO 2 — Restoring Division (Simulation & Analysis)

■ Section C: Paper Simulation (Apply): 0110 (6) ÷ 0010 (2) — restoring division

- Init: $A = 0000$, $Q = 0110$, $M = 0010$
- Step 1: shift (A,Q) → $A = 0000$, $Q = 1100$; $A = A - M = 1110 (< 0)$ → restore $A = 0000$, $Q_0 = 0$ → $A = 0000$, $Q = 1100$
- Step 2: shift → $A = 0001$, $Q = 1000$; $A = A - M = 1111 (< 0)$ → restore $A = 0001$, $Q_0 = 0$ → $A = 0001$, $Q = 1000$
- Step 3: shift → $A = 0011$, $Q = 0000$; $A = A - M = 0001 (\geq 0)$ → $Q_0 = 1$ → $A = 0001$, $Q = 0001$
- Step 4: shift → $A = 0010$, $Q = 0010$; $A = A - M = 0000 (\geq 0)$ → $Q_0 = 1$ → $A = 0000$, $Q = 0011$
- Final Quotient = 0011 (3), Remainder = 0000 (0) ($6 \div 2 = 3$ remainder 0 ✓)

■ Section D: Role-Based Play Debrief (Analyze)

- Restoration is triggered whenever the subtraction produces a negative remainder — the subtraction 'failed', so the divisor is added back to recover the previous remainder.
- The division is complete after n cycles, where n is the number of bits of the quotient/dividend register; the quotient sits in Q and the remainder in A .
- Each left shift brings the next dividend bit into the remainder for the trial subtraction and simultaneously frees the LSB of Q to receive the new quotient bit.